

MongoDB Query Optimization Report

Execution Plan — Before Indexing

```
bookstore> // Task 4. MongoDB Query Optimization -----
| db.books.find(
|   category: "Programming",
|   published_year: { $gte: 2020 }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookstore.books',
    parsedQuery: {
      '$and': [
        { category: { '$eq': 'Programming' } },
        { published_year: { '$gte': 2020 } }
      ]
    },
    indexFilterSet: false,
    queryHash: '702E1AA4',
    planCacheShapeHash: '702E1AA4',
    planCacheKey: '57E0A254',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        '$and': [
          { category: { '$eq': 'Programming' } },
          { published_year: { '$gte': 2020 } }
        ]
      },
      nss: 'bookstore.books',
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 0,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 33,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        '$and': [
          { category: { '$eq': 'Programming' } },
          { published_year: { '$gte': 2020 } }
        ]
      }
    }
  }
}
```

```

    { category: { '$eq': 'Programming' } } },
    { published_year: { '$gte': 2020 } } }
  ]
},
nReturned: 0,
executionTimeMillisEstimate: 0,
works: 34,
advanced: 0,
needTime: 33,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'bookstore.books',
direction: 'forward',
docsExamined: 33
}
},
queryShapeHash: '645208f9871e861ace7ff0d082ac0305644ac41e7868cf04ab331bf413c8cea4',
command: {
  find: 'books',
  filter: { category: 'Programming', published_year: { '$gte': 2020 } },
  '$db': 'bookstore'
},
serverInfo: {
  host: 'DESKTOP-S0V7NKS',
  port: 27017,
  version: '8.3.4',
  gitVersion: '4b03e7daaa316c78b9bf433046dba81637d581c0'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

Execution Plan — After Indexing

```
bookstore> db.books.createIndex({
|   category: 1,
|   published_year: 1
| })
category_1_published_year_1
bookstore> db.books.find({
|   category: "Programming",
|   published_year: { $gte: 2020 }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookstore.books',
    parsedQuery: {
      '$and': [
        { category: { '$eq': 'Programming' } },
        { published_year: { '$gte': 2020 } }
      ]
    },
    indexFilterSet: false,
    queryHash: '702E1AA4',
    planCacheShapeHash: '702E1AA4',
    planCacheKey: '4C5FF9D1',
    optimizationTimeMillis: 21,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'bookstore.books',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'bookstore.books',
        keyPattern: { category: 1, published_year: 1 },
        indexName: 'category_1_published_year_1',
        isMultiKey: false,
        multiKeyPaths: { category: [], published_year: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          category: [ '['Programming', 'Programming'] ' ],
          published_year: [ '[2020, inf)' ]
        }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
```

```

nReturned: 0,
executionTimeMillis: 26,
totalKeysExamined: 0,
totalDocsExamined: 0,
executionStages: {
  isCached: false,
  stage: 'FETCH',
  nReturned: 0,
  executionTimeMillisEstimate: 0,
  works: 1,
  advanced: 0,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'bookstore.books',
  docsExamined: 0,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 0,
    executionTimeMillisEstimate: 0,
    works: 1,
    advanced: 0,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookstore.books',
    keyPattern: { category: 1, published_year: 1 },
    indexName: 'category_1_published_year_1',
    isMultiKey: false,
    multiKeyPaths: { category: [], published_year: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      category: [ '['Programming', 'Programming']' ],
      published_year: [ '[2020, inf]' ]
    },
    keysExamined: 0,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
},
queryShapeHash: '645208f9871e861ace7ff00d82ac0305644ac41e7b68cf04a83318f413c8cea4',

```

```
queryShapeHash: '645268FY87IE861ALE7FF000B2AL8305644AL41E7868CF04AB331BF413LBLEK4',
command: {
  find: 'books',
  filter: { category: 'Programming', published_year: { '$gte': 2020 } },
  '$db': 'bookstore'
},
serverInfo: {
  host: 'DESKTOP-S0V7NKS',
  port: 27017,
  version: '8.3.4',
  gitVersion: '4b03e7daaa316c78b9bf433046dba81637d581c0'
},
serverParameters: {
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  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
```

Execution Statistics Summary

Metric	Before Index (COLLSCAN)	After Index (IXSCAN)
Scann performed	COLLSCAN	FETCH → IXSCAN
Total documents examined (totalDocsExamined)	33	0 (no docs fitted for filter)
Execution time (executionTimeMillis)	0 ms	1 ms

Analysis

1. Why do indexes improve performance?

- Instead of looking at all values in table machine is able to quickly sort through indexes

2. What are the differences between COLLSCAN and IXSCAN?

- **COLLSCAN** – walk through every document + check filter linearly
- **IXSCAN** – refers to stored values as compound index directly, no check of unnecessary documents

3. How many documents were examined before and after indexing?

- 33 → 0

4. Was the execution time improved? Explain the result.

- 0 ms → 1 ms ⇒ not improved as for such a small DB it is faster to look through ~30 books than to use indexes (jump to needed category → needed years → fetch)